

Smart Traffic Violation Detection Report

Project: Traffic Violation Detection System

Date: May 30, 2026

Implemented features:

- Helmet violation detection for two-wheelers
- Phone-in-hand rider detection
- Triple-seat rider detection
- Live YOLOv8 processing and DeepSORT tracking
- Violations saved with evidence images
- Owner SMS notification via number plate registry

Updated dashboard:

- Added Phone Violations and Triple Seat Violations metrics

- Live processed feed and latest violation card

How to run:

1. Activate .venv

2. Install dependencies: `pip install -r requirements.txt`

3. Start server: `uvicorn backend.main:app --host 0.0.0.0 --port 8000`

4. Open `http://127.0.0.1:8000`

Important files:

- backend/detection.py

- backend/database.py

- frontend/index.html

- frontend/app.js

- frontend/styles.css

- docs/ui_overview.png

- Traffic_Violation_Report.pdf

Notes:

The system now detects helmet, phone use, and triple-seat violations in real time from the camera feed.

Violations are stored in SQLite and shown on the dashboard with image evidence.

UI Overview Screenshot

